

OmniSight

AI-Driven Predictive Maintenance & AR Digital Twin System

PROJECT SYNOPSIS

GUIDE

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TEAM

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AI

Abstract

- A low-cost, AI-driven predictive maintenance system paired with an Augmented Reality (AR) Digital Twin, built to bring Industry 4.0-grade diagnostics within reach of MSME budgets.
- Sensing — an ESP32 edge microcontroller with an MPU6050 sensor mounted on machinery, continuously capturing tri-axial vibration and thermal telemetry.
- AI Analysis — a Random Forest model analyses eight time-domain features (crest factor, peak shock, and others) to classify faults such as bearing wear and shaft misalignment.
- AR Interface — a Unity + Vuforia “point-and-see” overlay shows live, colour-coded diagnostic tags directly on the machine.
- Designed as a self-built prototype, targeted at a total cost under ₹30,000.

SYSTEM STACK

01 Sense

ESP32 + MPU6050

Vibration & thermal capture

02 Think

Random Forest ML

8 time-domain features

03 See

Unity + Vuforia AR

Colour-coded diagnostic tags

Introduction

Domain

Industrial IoT and Edge AI applied to predictive maintenance, visualised through Augmented Reality.

Motivation

MSMEs rely on reactive, break-then-fix maintenance because enterprise-grade Industry 4.0 monitoring tools are cost-prohibitive at their scale.

Context

Built for small and medium manufacturing setups — job shops, fabrication units and local factories — that cannot justify large monitoring-system budgets.

Approach

Combine affordable edge hardware (ESP32 + MPU6050), an on-device-trainable ML model (Random Forest), and an immersive AR interface into one deployable system.

WHY AR?

Shop-floor workers rarely read raw sensor dashboards. OmniSight overlays a colour-coded health tag directly on the machine through a phone camera — green, amber, or red — so a diagnosis is understood in seconds, without training.

Problem Statement

1

MSMEs depend on reactive maintenance — machines are repaired only after they fail.

2

Unplanned downtime, high repair costs, and safety risks follow every unexpected breakdown.

3

Industry 4.0 predictive-maintenance platforms exist, but their cost puts them out of reach for small manufacturers.

4

A low-cost, AI-driven system is needed to bring proactive, visual fault detection within MSME budgets.

Net effect: machines fail without warning, and MSMEs absorb the downtime, repair, and safety cost that early detection could have prevented.

Need for the Project

- Enterprise predictive-maintenance platforms are built for large factories — none are priced or packaged for an MSME workshop.
- There is a real gap between costly, dashboard-heavy Industry 4.0 systems and a tool a small manufacturer can actually afford and use.
- Vibration and thermal degradation in bearings and shafts give early warning signs weeks before failure — signals that go unmonitored today.
- A sub-₹30,000 system built on commodity sensors and an intuitive AR overlay does not currently exist for this segment.
- A Random Forest model trained on time-domain features offers reliable, retrainable fault classification without deep-learning-scale compute.

THE GAP

**Enterprise Predictive
Maintenance Platforms**



**GAP: no affordable,
AR-guided fallback for MSMEs**



**OmniSight
(this project)**

Objectives

GENERAL OBJECTIVE

To design a low-cost, AI-driven predictive maintenance system with an AR Digital Twin interface that detects early mechanical faults in MSME machinery and visualises them in real time, within a build cost under ₹30,000.

SPECIFIC OBJECTIVES

01 Capture tri-axial vibration and thermal telemetry using an ESP32 + MPU6050 sensor node

02 Stream telemetry to a Python backend via MQTT over factory Wi-Fi with low latency

03 Extract eight time-domain features (crest factor, peak shock, and others) from raw signals

04 Train a Random Forest model to classify faults such as bearing wear and shaft misalignment

05 Build a Unity + Vuforia AR interface showing live, colour-coded diagnostic tags on-machine

06 Maintain historical asset health records to support trend-based decision making

Existing System

Current MSME maintenance relies on reactive repair, manual inspection rounds, and — where affordable at all — enterprise Industry 4.0 monitoring dashboards designed for large factories.

LIMITATIONS	ADVANTAGE
Reactive maintenance — fixed only after failure	Proactive, AI-driven fault detection pre-breakdown
Enterprise tools cost-prohibitive for MSMEs	Affordable hardware and build — under ₹30,000
Complex dashboards need trained personnel	Intuitive AR — point a phone and see the diagnosis
No real-time visual diagnostic aid on the floor	Real-time MQTT telemetry with low latency
Limited scalability across older machine types	Retrainable model, scalable to new machine types

Proposed System

A four-layer architecture converging sensing, AI decisioning, telemetry and AR visualisation into one system that serves a technician's phone directly.

01

DATA COLLECTION**ESP32 + MPU6050**

Tri-axial vibration & thermal capture

02

AI DECISION**Random Forest (Python)**

8 time-domain features, fault classification

03

TELEMETRY & STORAGE**MQTT + historical log**

Low-latency streaming, asset records

04

USER INTERFACE**Unity + Vuforia AR**

Colour-coded diagnostic tags, hazard overlay

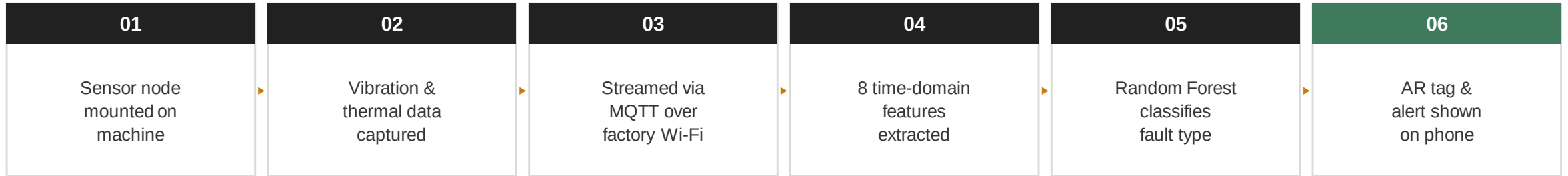
▼ Converges into

OMNISIGHT — CONVERGED SYSTEM → Technician's Phone (AR view)**MAJOR COMPONENTS**

- ESP32 edge controller
- MPU6050 IMU sensor
- MQTT broker (Wi-Fi)
- Python + scikit-learn
- Random Forest classifier
- Unity + Vuforia app

Methodology

END-TO-END MONITORING FLOW



AI FAULT-CLASSIFICATION PIPELINE



Model accuracy depends on the quality and diversity of training vibration data collected across machine conditions before deployment.

Tools & Technologies

HARDWARE

ESP32 Edge Controller

MPU6050 IMU Sensor

Industrial Mounting Bracket

Wi-Fi Connectivity Module

SOFTWARE / AI

Unity Engine

Vuforia AR SDK

scikit-learn (Random Forest)

Mosquitto MQTT Broker

PROGRAMMING LANGUAGES

Python

C / C++ (ESP32 firmware)

C# (Unity scripts)

DATA / STORAGE

Telemetry Log (CSV / SQLite)

Historical Asset Records

Trained Model File (.pkl)

No conventional relational database is mandated — lightweight structured logs (CSV/SQLite) hold telemetry and asset history. Final storage-service choice is a reasonable engineering default, pending confirmation.

Output of First Two Modules

MODULE 01

SENSOR TELEMETRY PIPELINE

ESP32 + MPU6050 node capturing tri-axial vibration and thermal readings, streamed live over MQTT to the Python backend with no external cloud dependency.

- Sensor → MQTT → backend link established
- Continuous vibration & thermal stream verified
- Low-latency delivery over factory Wi-Fi



Live Telemetry Stream

Screenshot / demo output to be inserted at review time

MODULE 02

AI FAULT CLASSIFICATION ENGINE

A Random Forest model trained on eight time-domain features to distinguish healthy operation from early bearing wear and shaft misalignment.

- Feature extraction pipeline working end-to-end
- Baseline vs. faulty signal classification tested
- Model exportable for on-device inference



Fault Classification Report

Screenshot / demo output to be inserted at review time

Expected Outcomes of the Complete Mini Project

- 01 A fully functional low-cost predictive maintenance prototype, built for under ₹30,000
- 02 A working AR interface (Unity + Vuforia) displaying live, colour-coded diagnostic tags on-machine
- 03 A trained Random Forest model reliably classifying bearing wear and shaft misalignment in real time
- 04 A low-latency MQTT telemetry pipeline streaming vibration and thermal data with zero appreciable lag
- 05 Historical asset health records enabling trend-based maintenance decisions over time
- 06 A reusable framework extensible to CNCs, conveyors, HVAC systems, and packaging lines

Project Timeline



Future Scope

NEAR-TERM (POST-PROTOTYPE)

- Expand fault classification to more machine and bearing types
- Field testing on live MSME shop floors under real operating loads
- Package the AR app for easy install on any Android device
- Add a maintenance-manager dashboard with trend logs and alerts

LONGER-TERM VISION

- Multi-machine, factory-wide monitoring network
- Predictive-maintenance-as-a-service for MSME clusters
- Integration with ERP / MES systems for automated work orders
- Cloud-optional pipeline for periodic model retraining

References

- L. Breiman, “Random Forests,” Machine Learning, 2001.
- Espressif Systems, “ESP32 Series Technical Reference Manual.”
- InvenSense, “MPU-6050 Product Specification.”
- OASIS, “MQTT Version 3.1.1 Protocol Specification.”
- Unity Technologies, Official Documentation, unity.com.
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- R. B. Randall, “Vibration-based Condition Monitoring,” Wiley, 2011.
- Eclipse Mosquitto Project, Official Documentation, mosquitto.org.

Representative academic and technical references for the technologies used; no reference list was supplied in the source project document.

Thank You

Questions?

OMNISIGHT: AI-DRIVEN PREDICTIVE MAINTENANCE & AR DIGITAL TWIN SYSTEM

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